# JOEL THOMAS CYRIL

**Electronics and Computer Engineering Applicant | Embedded Systems | Full-Stack Development**

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## PROFILE

CBSE Class XII student pursuing undergraduate study in Electronics and Computer Engineering, with hands-on experience designing custom electronic devices, developing embedded firmware, and building practical software systems. Projects span ESP32 and STM32 hardware, RF and LoRa communications, robotics, AI-assisted education tools, and self-hosted web applications.

## EDUCATION

**Abu Dhabi Indian School (ADIS)** — Abu Dhabi, UAE
CBSE Senior Secondary (Class XII) • Computer Science, Physics and Mathematics
CBSE Class X board results: **Mathematics 100/100** • **English Language and Literature 95/100**

**IIT Madras, Centre for Outreach and Digital Education (CODE)** — 2026
Introduction to Electronic Systems • Eight-week certificate course in circuits, semiconductors, op-amps and digital electronics

## TECHNICAL SKILLS

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|---|---|
| **Languages** | C, C++, Python, TypeScript, JavaScript, SQL, HTML and CSS |
| **Embedded and Control** | ESP32, STM32WLE5, FreeRTOS, PID control, sensors |
| **PCB and Interfaces** | KiCad, PCB design, I2C, SPI, UART, RS-485 |
| **Wireless and RF** | LoRa, CC1101 Sub-GHz RF, PN532 NFC/RFID, infrared |
| **Web and Backend** | React, Node.js, FastAPI, REST APIs, WebSockets, Vite, Tailwind CSS |
| **Data** | PostgreSQL, SQLite, Supabase, Tesseract OCR |
| **Tools and Infrastructure** | Linux, Docker, Git/GitHub, Vercel |

## SELECTED ENGINEERING PROJECTS

**Reaper One** | Multi-Protocol Handheld Device — 2024–2026
**ESP32 • C++ • KiCad • FreeRTOS • CC1101 • NFC/RFID** | github.com/JoelCyril/reaper1.2

- Designed and assembled a compact ESP32-based device integrating Sub-GHz radio, NFC/RFID, infrared, Wi-Fi and Bluetooth LE, with a custom circuit board and 3D-designed enclosure.
- Developed menu-driven C++ firmware and integrated modules over SPI, I2C and GPIO; iterated hardware and firmware prototypes over approximately 18 months.

**ADIS ReadTrack** | AI-Assisted Reading Assessment Platform — 2026
**React • TypeScript • Supabase • PostgreSQL • Tesseract OCR**

- Developed a school reading and assessment portal in consultation with English teachers, designed to support approximately 2,000 ADIS students.
- Implemented student-card OCR onboarding, reading activity tracking, AI-assisted assessments and class-level literacy leaderboards.

**SwapUAE** | Cashless Community Marketplace 2026

**React • TypeScript • Node.js • PostgreSQL • WebSockets** | github.com/JoelCyril/swap

- Co-developed a peer-to-peer barter platform with listings, trade offers and real-time messaging; early launch reached 100+ users, 110+ listings and 12+ trades.
- Built a self-hosted Linux NAS from reused computer parts to support affordable application infrastructure after free cloud limits became restrictive.

**Manarat Al-Qurm** | Mangrove Monitoring and LoRa Telemetry 2026

**STM32WLE5 • LoRa • Raspberry Pi • Python • FastAPI • SQLite**

- Designed a low-power coastal monitoring architecture transmitting sensor readings from a buoy to a local Raspberry Pi gateway using LoRa.
- Developed a dashboard for six water-quality parameters and offline 48-hour trend forecasting to support environmental monitoring.

**Underwater Remotely Operated Vehicle (ROV)** | Robotics 2026

**ESP32 • C++ • RS-485 • Brushless motors • PID control**

- Developed an underwater ROV using a tethered communication link, brushless thrusters, depth and motion sensors, and PID-based stabilization.

**Pulso.ai** | AI Wellness Analytics Platform 2026

**React • TypeScript • Vite • Data visualization** | github.com/JoelCyril/Pulso.ai

- Built an interactive application synthesizing lifestyle inputs into wellness insights and visual dashboards aligned with UN Sustainable Development Goal 3.

**Home NAS** | Self-Hosted Infrastructure 2026

**Linux • PostgreSQL • Docker • ZFS/RAID • Nginx**

- Reused discarded computer components to configure local data storage, backups and reverse-proxy services for community applications, reducing dependence on paid hosting.

## LEADERSHIP AND EXTRACURRICULAR ACTIVITIES

**Secretary, ADIS Robotics Club** 2023–Present

Coordinated workshops and lab activities; mentored junior members in EV3/SPIKE programming and autonomous robotics.

**Team Captain, FIRST LEGO League UNEARTHED** 2025–2026

Led robot design and mission strategy; team placed First Runner-Up in the regional Robot Game.

**Varsity Football, ADIS** 2024–2026

Represented the school at the ADEK Cup and the CBSE Cluster U-17 tournament.

**Model United Nations Delegate** 2024–2025

Participated in committee debate, research and resolution drafting.

## SELECTED AWARDS AND RECOGNITION

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|---|---|---|
| 2026 | **RIT Annual Engineering Competition** | First Runner-Up; AED 20,000 scholarship |
| 2026 | **ADU STEM Engineering and Coding Competition** | Top 10 of 512 teams (2,083 students, 92 schools) |
| 2026 | **Curtin University Dubai STEM Challenge** | Top 12 finalist team, ADIS Elites |
| 2025 | **Inter-School English Short Film** | Second position, Indian School Al Ain |
| 2024 | **Mayoor Inter-School STEM Competition** | First Runner-Up, senior division |
| 2022–2025 | **Chess and Speedcubing** | 1st Marthoma Smruthi Chess (2025); 1st ADIS Speedcubing (2022); 2nd School Chess (2024) |
| 2024–2025 | **Football and Other Sports** | 1st ADIS Inter-House Football (2024); Badminton runner-up (2025); Carrom doubles 2nd (2025) |
| 2023–2024 | **Performing Arts** | First prize, Malayalam solo singing |